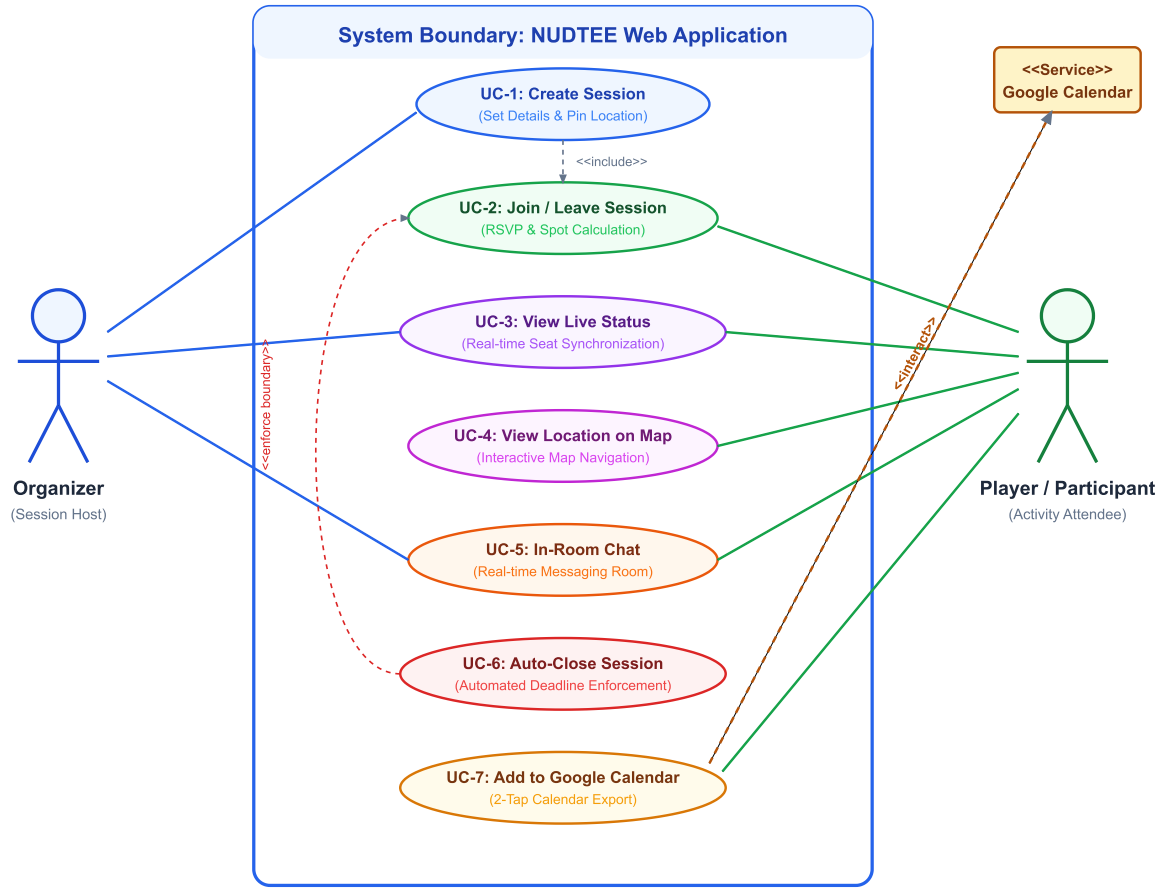
NUDTEE Use Case Diagram

System functionality and interaction model for real-time activity scheduling

Overview: This Use Case Diagram illustrates the three core sections of the NUDTEE platform: **Organizer (Left Actor)** who creates and manages activity sessions, **System Boundary (Center)** encompassing functional requirements FR-1 through FR-7 with dynamic system automation, and **Player / Participant (Right Actor)** alongside external services like Google Calendar.



Functional Requirements (FR) Traceability Matrix

FR Code	Use Case Name	Primary Actor	Key Operations, Business Logic & Constraints
FR-1	UC-1: Create Session	Organizer	Creates an activity with title, min/max players, date/time, deadline, category, map location or online link. Organizer is automatically joined as the first participant.
FR-2	UC-2: Join / Leave Session	Player	Join or leave in 2 taps or fewer. Uses Cloud Firestore Transactions to prevent race conditions (seat overbooking) and checks for time overlaps with existing joined rooms.
FR-3	UC-3: View Live Status	Organizer / Player	Real-time synchronization of participant counts and room statuses (Open, Almost Full, Full, Joined, Closed) within 2 seconds without requiring manual page refresh.
FR-4	UC-4: View Location on Map	Player	Displays interactive map centered on venue coordinates with pan/zoom controls and an "Open in Google Maps" link. Online sessions display meeting links instead.

FR Code	Use Case Name	Primary Actor	Key Operations, Business Logic & Constraints
FR-5	UC-5: In-Room Chat	Organizer / Player	Real-time dedicated chat room accessible exclusively to joined participants for logistical coordination. Becomes read-only after session completion.
FR-6	UC-6: Auto-Close Session	System Logic	Automatically compares current time against registration deadline (23:59) or start time. Changes status to "Closed" and rejects 100% of late join attempts.
FR-7	UC-7: Add to Google Calendar	Organizer / Player	Exports session schedule and location to Google Calendar via pre-filled template link in 2 taps or fewer to set reminders and prevent forgotten appointments.